

Chapter 1: Vectors

A vector is a quantity with both magnitude and direction.

We write vectors in bold, such as $\mathbf{v}$, to distinguish them from scalars.

Vector addition combines two vectors tip to tail.

The result is a new vector from the start of the first to the end of the second.



Chapter 2: Matrices

A matrix is a rectangular array of numbers arranged in rows and columns.

Matrices can represent linear transformations of vectors.

Matrix multiplication combines two matrices by rows and columns.

The number of columns in the first matrix must match the rows in the second.

Chapter 3: Determinants

The determinant of a matrix is a single number computed from its entries.

A zero determinant means the matrix has no inverse.